2025 AMC 10A Problem 11 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 11. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 11

Problem:
The sequence $1, x, y, z$ is arithmetic. The sequence $1, p, q, z$ is geometric. Both sequences are strictly increasing and contain only integers, and z is as small as possible. What is the value of $x + y + z + p + q$?

Answer Choices:

A. 66

B. 91

C. 103

D. 132

E. 149

Solution:

Let $1, x, y, z$ be an arithmetic sequence and $1, p, q, z$ be a geometric sequence of strictly increasing positive integers.

If the common difference of the arithmetic sequence is $d > 0$, then

$$x = 1 + d, \quad y = 1 + 2d, \quad z = 1 + 3d.$$

If the common ratio of the geometric sequence is $r > 1$, then

$$p = r, \quad q = r^2, \quad z = r^3.$$

Thus,

$$1 + 3d = r^3.$$

Since d and r are positive integers, $r^3 - 1$ must be divisible by 3.

This happens whenever r leaves a remainder 1 upon division by 3.

The smallest integer $r > 1$ with this property is $r = 4$.

Hence,

$$z = r^3 = 64, \qquad d = \frac{64 - 1}{3} = 21.$$

The arithmetic sequence is

$$1,\ 22,\ 43,\ 64,$$

and the geometric sequence is

$$1,\ 4,\ 16,\ 64.$$

Both are strictly increasing sequences of positive integers, and $z = 64$ is minimal.

Finally,

$$x + y + z + p + q = 22 + 43 + 64 + 4 + 16 = \boxed{\textbf{(E) } 149}.$$